6.2202 class notes

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1 Circuit review

Laws. KCL: at every node, $\sum$(currents in) = $\sum$(currents out). KVL: around every closed loop, $\sum V = 0$.

Phasors. Sinusoid $v(t) = V_{max}\cos(\omega t + \phi)$. Positive ϕ shifts the curve left, so the waveform occurs earlier. RMS value: $\tilde{V} = V_{rms} = V_{max}/\sqrt{2}$. Power engineering reports V_{rms}, not V_{max}; state which you use. The phasor of $v(t) = \mathrm{Re}\{\tilde{V}\cos(\omega t+\phi) + j\tilde{V}\sin(\omega t+\phi)\} = \mathrm{Re}\{\tilde{V}e^{\omega t+\phi}\}$. Write as phasor: $V = \tilde{V}\angle\phi$, a complex number with RMS magnitude and angle ϕ. $j = \sqrt{-1} = 1\angle 90°$, and $1/j = -j$. Rectangular $\iff$ polar: $a + jb = \sqrt{a^2+b^2}\angle\,\mathrm{atan2}(b,a)$, and $r\angle\theta = r\cos\theta + jr\sin\theta = re^{j\theta}$. Add and subtract in rectangular. Multiply and divide in polar: $(r_1\angle\theta_1)(r_2\angle\theta_2) = r_1r_2\angle(\theta_1+\theta_2)$, $(r_1\angle\theta_1)/(r_2\angle\theta_2) = (r_1/r_2)\angle(\theta_1-\theta_2)$. Conjugate: $(r\angle\theta)^* = r\angle(-\theta)$.

Elements. Sign convention: I is defined entering the + terminal of V. Impedance $Z = R + jX$ [Ω], with resistance R and reactance X. Polar: $Z = |Z|\angle\theta$, $|Z| = \sqrt{R^2+X^2}$, $\tan\theta = X/R$. Admittance $Y = 1/Z = G + jB$ [S], with conductance G and susceptance B; in polar $Y = (1/|Z|)\angle(-\theta)$. Ohm's law: $V = IZ$, or $I = VY$. With $V = \tilde{V}\angle\phi_V$ and $I = \tilde{I}\angle\phi_I$, $\theta = \phi_V - \phi_I$. Inductor L: $v = L\,di/dt$, so $X_L = \omega L$. Capacitor C: $i = C\,dv/dt$, so $X_C = -1/(\omega C)$. Inductive ("lag"): $\theta > 0$, $\phi_I < \phi_V$, current occurs after voltage. Capacitive ("lead"): $\theta < 0$, $\phi_I > \phi_V$, current occurs before voltage.

Solving. Series: $Z_{eq} = Z_1 + Z_2 + \ldots$; same current through each. Parallel: $Y_{eq} = Y_1 + Y_2 + \ldots$, or $Z_{eq} = Z_1Z_2/(Z_1+Z_2)$ for two; same voltage across each. Voltage divider (series): $V_2 = V \cdot Z_2/(Z_1+Z_2)$. Current divider (parallel): $I_2 = I \cdot Z_1/(Z_1+Z_2)$. Thevenin: a source V_{Th} in series with Z_{Th}. V_{Th} is the open-circuit voltage at the terminals. Z_{Th} is the impedance seen at the terminals with independent sources zeroed (voltage source → short, current source → open), or $Z_{Th} = V_{oc}/I_{sc}$. Norton: a current source $I_N = V_{Th}/Z_{Th}$ in parallel with $Z_N = Z_{Th}$.

Transients. First-order RC or RL circuit with a DC source: $x(t \geq 0) = x(\infty) + [x(0^+) - x(\infty)]\,e^{-t/\tau}$ for any voltage or current x. Time constant $\tau = R_{Th}C$ (RC) or L/R_{Th} (RL), with R_{Th} the Thevenin resistance seen by v_C or i_L. v_C and i_L cannot jump: $x(0^+) = x(0^-)$. $x(\infty)$: capacitor open, inductor short. Any other variable at 0^+: redraw with C as a voltage source $v_C(0^+)$ (or L as a current source $i_L(0^+)$) and solve. Origin: particular solution (steady state) plus homogeneous $Ae^{-t/\tau}$, with A set by $x(0^+)$. Power absorbed by any element: $p(t) = v(t)\,i(t)$. Stored energy is $w(t) = \int_0^t p(t)\,dt = \int v(t)\,i(t)\,dt$. Capacitor: $i(t)\,dt = C\,dv$, so $w(t) = \int v(t)\,C\,dv = \frac{1}{2}Cv(t)^2$. Inductor: $v(t)\,dt = L\,di$, so $w(t) = \int L\,i(t)\,di = \frac{1}{2}Li(t)^2$.

2 Power

Definitions. Complex power $S = VI^*$ [VA]; S is power absorbed by the element. $S = P + jQ$, with real power P [W] and reactive power Q [VAr]. Polar: $S = \tilde{V}\tilde{I}\angle\theta$, where $\theta = \phi_V - \phi_I$. So $P = \tilde{V}\tilde{I}\cos\theta$, $Q = \tilde{V}\tilde{I}\sin\theta$, $|S| = \tilde{V}\tilde{I}$, and $Q = P\tan\theta$. Power factor pf $= \cos\theta = P/|S|$. $\cos\theta$ loses the sign of θ, so a pf is stated lagging or leading. Lagging: $\theta > 0$, inductive, $Q > 0$, absorbs Q. Leading: $\theta < 0$, capacitive, $Q < 0$, supplies Q. Single element: $S = \tilde{I}^2 Z = \tilde{V}^2 Y^*$.

Derivation of P. $p(t) = v(t)\,i(t) = V_{max}I_{max}\cos(\omega t + \phi_V)\cos(\omega t + \phi_I)$. By $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$, $p(t) = \frac{1}{2}V_{max}I_{max}[\cos(\phi_V - \phi_I) + \cos(2\omega t + \phi_V + \phi_I)]$. The first term is constant; the second averages to zero. So $P = \frac{1}{2}V_{max}I_{max}\cos\theta = \tilde{V}\tilde{I}\cos\theta$.

Load to element. A problem states a load as $|S|$ (or P) and a pf, lagging or leading. $\theta = \pm\arccos(\text{pf})$, positive if lagging. As a current: $I = (S/V)^*$, so $\tilde{I} = |S|/\tilde{V}$ and $\phi_I = \phi_V - \theta$. As an impedance: $Z = \tilde{V}^2/S^*$.

Conservation. $\sum S = 0$ at a node and P and Q balance separately. Power-factor correction: given P, pf θ_1, bus $V = \tilde{V}\angle 0$, target θ_2; find C in parallel with the load. $S_L = P + jP\tan\theta_1$; $S_C = |V|^2 Y^* = -j\omega C\tilde{V}^2$; $S_{src} = S_L + S_C = P + j(P\tan\theta_1 - \omega C\tilde{V}^2)$. Target: $Q_{src} = P\tan\theta_2$, since $Q = P\tan\theta$ and P is unchanged.

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